EAST WEST UNIVERSITY

Department of Computer Science and Engineering

B.Sc. in Computer Science and Engineering Program

Final Examination, Summer 2025

Course: CSE 106 Discrete Mathematics, Section-03 **Date:** September 10, 2025
Instructor: Dr. Nafis Sadeq, Assistant Professor, CSE Department
Full Marks: 45
Time: 1 hour and 10 minutes

Note: There are **TEN** questions. Answer **ALL** of them. Course Outcome (CO), Cognitive Level, and Mark of each question are mentioned at the right margin.

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1. How many binary strings of length 12 either start with three ones or end with two zeros?

[CO2, C2] [Mark: 5]

2. Each number in the array A is divided by 7. Determine the **minimum size** of the array such that **at least 12 numbers have the same remainder** modulo 7.

[CO2, C2] [Mark: 5]

3. Let x, y be two positive integers. Their prime factorization is given by:
$x = 2^4 \times 5^3 \times 11^2 \times 17, \quad y = 3^4 \times 2^3 \times 7^3.$

(a) How many unique positive integers can divide x?

(b) How many unique positive integers can divide both x and y?

[CO2, C3] [Mark: 5]

4. Determine the total number of executed instructions and worst-case time complexity for searching a key k in an unsorted array A of size n. The pseudocode is given below.

Algorithm 1 Linear Search in Array A

```
1: location ← −1
2: for i ← 1 to n do
3:     if A[i] = k then
4:         location ← i
5:         break
6:     end if
7: end for
```

[CO3, C2] [Mark: 6]

5. Determine the **tightest** big-O estimate for each of the following functions.

(a) $(n^2 + n \log n)(\log n + 5) + (9n^3 + 4 \log n)$

(b) $(3^n + n^5)(2^n + n^2)$

(c) $(n^{2n} + 4^n)(n! + n^n)$

[CO3, C3] [Mark: 6]

6. Find the following modular computations:

(a) $\left(41^3 \bmod 17\right) \bmod 6$

(b) $(27^2 \bmod 19)^3 \bmod 11$

[CO3, C3] [Mark: 5]

7. Determine if the Graph $G2$ shown in Figure 1 is a cycle. Justify your answer.

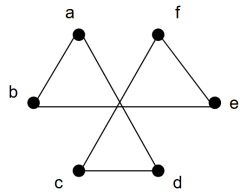


Figure 1: Graph $G2$

[CO4, C2] [Mark: 3]

8. Determine whether Graph $G2$ in Figure 1 is bipartite. Justify your answer.

[CO4, C3] [Mark: 3]

9. Determine the **level** of every node in the tree shown in Figure 2.

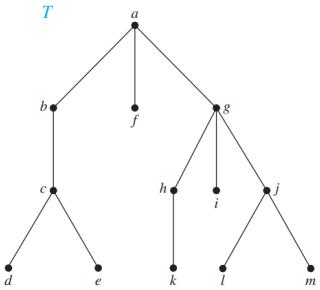


Figure 2: A Tree

[CO4, C2] [Mark: 3]

10. Given a full 6-ary tree with 5 internal nodes, determine the number of leaf nodes.

[CO4, C3] [Mark: 4]

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Final Examination, Summer 2025

Course: CSE 106 Discrete Mathematics, Section-09 **Date:** September 9, 2025
Instructor: Dr. Nafis Sadeq, Assistant Professor, CSE Department
Full Marks: 45
Time: 1 hour and 10 minutes

Note: There are **TEN** questions. Answer **ALL** of them. Course Outcome (CO), Cognitive Level, and Mark of each question are mentioned at the right margin.

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1. A license plate consists of 7 characters, where each character is either a capital English letter (A–Z) or a digit (0–9). How many such plates either start with 4 digits or end with 4 digits?

[CO2, C2] [Mark: 5]

2. How many numbers must be selected from the set $\{1, 3, 5, 7, 9, 11, 13, 15\}$ to guarantee that at least one pair of these numbers add up to 16?

[CO2, C2] [Mark: 5]

3. A drawer contains 15 pairs of red, 10 pairs of blue, and 12 pairs of green socks. (a) What is the minimum number of individual socks one must pick to guarantee at least 5 socks of the same color? (b) If each pair contains a left-foot sock and a right-foot sock, what is the minimum number of socks needed to ensure at least 10 left-foot socks?

[CO2, C3] [Mark: 5]

4. Determine the total number of executed instructions and time complexity for computing the sum of all elements in an $m \times n$ matrix A. The pseudocode is given below.

Algorithm 1 Compute Sum of All Elements in an $m \times n$ Matrix

```
1: sum ← 0
2: for i ← 1 to m do
3:     for j ← 1 to n do
4:         sum ← sum + A[i][j]
5:     end for
6: end for
```

[CO3, C2] [Mark: 6]

5. Give the tightest big-O estimate for each of the following functions.

(a) $\big(n^3 + 4n \log n\big)(\log n + 2) + \big(13 \log n + 7\big)(2n^3 + 5)$

(b) $\big(n^4 + 4^n\big)(n^2 + 5^n)$

(c) $\big(n^{2^n} + n^n + 7^n\big)(n! + 6^n)$

[CO3, C3] [Mark: 6]

6. Is 5738733 a prime number? If not, determine the prime factorization of 5738733.

[CO3, C3] [Mark: 5]

7. Determine if the graph $G1$ shown in Figure 1 is a cycle. Justify your answer.

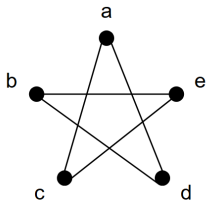


Figure 1: Graph $G1$

[CO4, C2] [Mark: 3]

8. Refer to the graph $G1$ in Figure 1. Determine whether it is bipartite with justification.

[CO4, C3] [Mark: 3]

9. Determine whether the undirected graph shown in Figure 2 is a tree. Justify your answer.

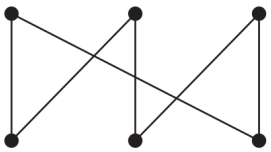


Figure 2: Undirected Graph

[CO4, C2] [Mark: 3]

10. Given a full 4-ary tree with 9 internal nodes, determine the total number of nodes.

[CO4, C3] [Mark: 4]

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Final Examination, Summer 2025

Course: CSE 106 Discrete Mathematics, Section-10 **Date:** September 9, 2025
Instructor: Dr. Nafis Sadeq, Assistant Professor, CSE Department
Full Marks: 45
Time: 1 hour and 10 minutes

Note: There are **TEN** questions. Answer **ALL** of them. Course Outcome (CO), Cognitive Level, and Mark of each question are mentioned at the right margin.

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1. How many positive integers less than 3000

(a) that are divisible by 13?

(b) that are divisible by 21?

(c) that are divisible by either 13 or 21?

[CO2, C2] [Mark: 5]

2. A drawer contains 20 red balls, 15 blue balls, and 10 green balls. What is the minimum number of balls that must be picked to guarantee that there are at least:

(a) 6 balls of the same color?

(b) 8 green balls?

[CO2, C2] [Mark: 5]

3. How many integers between 1 and 1000 (inclusive) are divisible by at least one of 3, 5, or 6?

[CO2, C3] [Mark: 5]

4. Determine the total number of executed instructions and time complexity for estimating the sum of the first n positive integers. The pseudocode is given below.

Algorithm 1 Sum of First n Positive Integers

```
1: sum ← 0
2: for i ← 1 to n do
3:     sum ← sum + i
4: end for
```

[CO3, C2] [Mark: 6]

5. Give a big-O estimate for each of these functions. For the function g in your estimate $f(x) = O(g(x))$, use a simple function g of smallest order.

(a) $(n^3 + n^2 \log n)(\log n + 1) + (17 \log n + 19)(n^3 + 2)$

(b) $(2^n + n^2)(n^3 + 3^n)$

(c) $(n^n + n^{2n} + 5^n)(n! + 5^n)$

[CO3, C3] [Mark: 6]

6. Compute the following modular expressions:

(a) $(19^2 \bmod 41) \bmod 9$

(b) $(32^3 \bmod 13)^2 \bmod 11$

[CO3, C3] [Mark: 5]

7. Find the in-degree and out-degree of each vertex in the graph G with directed edges shown in Figure 1.

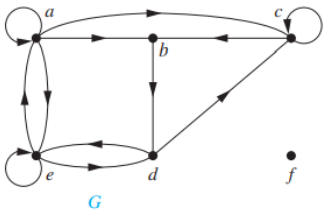


Figure 1: Directed graph G

[CO4, C2] [Mark: 3]

8. Compute the number of edges in a complete graph with 10 nodes.

[CO4, C3] [Mark: 3]

9. Determine whether the graph shown in Figure 2 is a tree. Justify your answer.

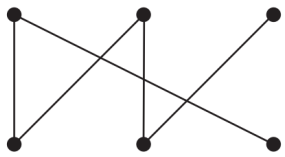


Figure 2: Undirected graph

[CO4, C2] [Mark: 3]

10. Determine the maximum number of leaves in a full 5-ary tree with height 4.

[CO4, C3] [Mark: 4]